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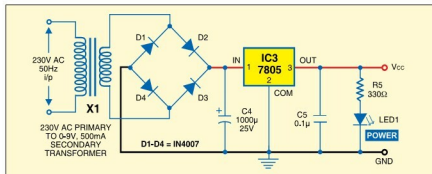


Fig. 3: Power supply circuit

PARTS LIST

Semiconductors:

- IC1 - ATmega16 AVR microcontroller
- IC2 - MAX6577 PWM temperature sensor
- IC3 - 7805, 5V regulator
- D1-D4 - 1N4007 rectifier diode
- LED1 - 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

- R1, R3 - 10-kilo-ohm
- R2 - 120-ohm
- R4 - 47-ohm
- R5 - 330-ohm
- VR1 - 10-kilo-ohm preset

Capacitors:

- C1 - 2.2μF, 16V electrolytic
- C2, C5 - 0.1μF ceramic disk
- C3 - 0.01μF ceramic disk
- C4 - 1000μF, 25V electrolytic

Miscellaneous:

- X1 - 230V AC primary to 9V, 500mA secondary transformer
- S1, S2 - Push-to-on switch
- LCD module (16×2)

the circuit is the AVR series microcontroller ATmega16 (IC1). The ATmega16 is a low-power CMOS, 8-bit microcontroller based on the RISC architecture. It has 16 kB of in-system programmable Flash memory with read-while-write capabilities, 512 bytes of EEPROM, 1kB SRAM, 32 general-purpose I/O lines and 32 general-purpose working registers.

The AVR core combines a rich instruction set with 32 general-purpose working registers. All the 32 registers are directly connected to the arithmetic logic unit (ALU), allowing two independent registers to be accessed in one single instruction executed in one

TABLE I
MAX6577 Time-Select

TS1	TS0	Scalar multiplier (Hz/°K)
GND	GND	4
GND	Vcc	1
Vcc	GND	1/4
Vcc	Vcc	1/16

clock cycle.

MAX6577 (IC2), used as the temperature sensor, converts the ambient temperature into a free-running, 50% duty-cycle square wave with a frequency that is proportional to the absolute temperature (°K) and is capable of measuring temperature in the range of -40°C to +125°C. Thus by measuring the output pulse frequency, the temperature can be calculated as follows:

$$T (^{\circ}\text{C}) = \text{Frequency(Hz)} / \text{Scalar Multiplier(Hz/}^{\circ}\text{K)} - 273.15^{\circ}\text{K}$$

The period/frequency range of the output square wave can be selected by hard-wiring the two time-select pins (TS0 and TS1) of MAX6577 to either Vcc or GND. Time-select pins (TS1 and TS0) permit the internal temperature-controlled oscillator (TCO) to be scaled by four preset multipliers.

The details of various ground and Vcc combinations of TS1 and TS0 are shown in Table I.

In the circuit (Fig. 2), MAX6577 has been rigged to generate output frequency with scalar multiplier as 1 (scalar multiplier in Hz/°K). For the purpose, pin TS0 is connected to Vcc and TS1 to ground.

The measurement accuracy of